MATH5007 2026 Semester 2: A1P1: Intro to Optimisation and AMPL

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Naming of Files: All files must start with your student ID, the semester, and A1P1 for Assignment 1 Part 1, e.g. "123456_S1_2026_A1P1" (called *FILE* in the following). Use this worksheet for your answer (use a different colour), rename it to *FILE*_answer.docx. AMPL files should have the format FILE.SUFFIX with SUFFIX being mod (model), dat (data), or run (run-script). If you submit multiple files, please create a zip file containing all files and name it *FILE*.zip.

Submission: TO BE ANNOUNCED ON BLACKBOARD

IMPORTANT: IN YOUR WORDS implies that you do not use the words from the lecture(r), slides, or online sources, but explain this in **YOUR OWN** words. Especially the use of AI for the final answers is not allowed.

Question 1: General (27 Marks)

The part is a short, general question for understanding and exploration.

1. What course are you enrolled in? Is this unit elective or mandatory for your course? **(2M)**

   I am enrolled in Master of Commerce, majoring in Supply Chain Management. This unit is a mandatory unit for my course.
2. This question is about your expectations on what you learn in this unit and if it helps to secure later an industry position **(5M)**

   From this unit, I am more interested to know the details about using the actual data to make better decisions in the context of supply chain management. I believe the unit is precisely based on analysing the actual present data, rather than using past data analytics in order to make the best supply chain decisions.

   The unit majorly focuses on how we can implement mathematics in supply chain analytics. The unit builds up my interest in knowing more on linear programming, max flow, routing and minimum spanning tree problems and utilise software like AMPL to solve them. Hence, mastering the network flow problems is what I would like to dig deeper on. Figuring out the most cost effective way to deal with transhipment problems, determining efficient routes for vehicles, and finding the cheapest way to connect points to best solve the minimum spanning tree problem is what I would like to learn about.

   Yes, I could see myself having a better chance of securing industry position specifically on how the unit is sub divided where it encourages student to build their communication techniques specifically via the oral exam, basically powering 50% of the weightage based on analytical ability and the other 50% is based on the student's communication ability.

3. Explain the following terms (**4 MARKS**):
   Model, Abstraction, Constraint, Infeasible Solution
   Use your own words and provide a short example.

   Model: A model is a structural representation of a real life problem or a system which allow to analyze the system or the problem, understand it and figure out the optimal solution without jeopardizing the real world results. Model is useful since the real world systems could be very complex hence, segregating it into just the necessary detail is the key reason why model is essential in supply chain.

Abstraction: An abstraction is the process of removing the unnecessary detail in the system or the problem and to focus on the relevant details which is an important segment of building a model. It enables to look into the core aspect without the complexity allowing the issues allowing to solve the issues mathematically.

Constraints: A constraint is the limitation in the model which enables them to know what preferred solution is considered feasible and acceptable. These constraints in a model should be real world scenarios such as the availability of resources, time restrictions, the budgets etc. which would ensure that the delegated solution is practically possible.

Infeasible solution: An infeasible solution is when the solution violates the constraints of the model which means it is not possible for it to be implemented in the real world scenario.

4. Write a short statement about AMPL and its application in the industry, using information on their Website and Google. What is your (educated) opinion of the product? **(6M)**

   AMPL which stands for A mathematical Programming Language, as the name suggests is a modelling programming language which is focused on solving the mathematical optimization problems. In the industry there are various companies which uses AMPL, one being Zara where it relies on AMPL to resolve their complex inventory problems where they got around 3000 items sold in 1500 stores enabling them to make decisions in no time after receiving the new data from sales (Caro et al. 2010). Similarly, Hitachi Energy uses AMPL in order to optimize their power grid management within their GridView platfom (Energy 2026).

5. Given are the inequalities 2x1+3x2<=10 and 3x1+5x2<=20. Visualise the solution area considering X1>=0 and X2>=0. Find the best solution assuming the objective function 15x1+10x2. Describe how you found the best solution visually <u>and</u> mathematically (under consideration, you know the visualisation of the solution space).
   Include an image (photo, scan, screenshot) of the graph in this document **(10M)**

   <u>First Constraint:</u>
   2x1 + 3x2 = 10, when x2= 0

   2x1=10

   Then x1= 10/2
   Hence x1=5

Therefore, the point is (5,0)

Now, when $x1= 0$

$3x2= 10$

therefore $x2= 10/3$

Second Constraint:

$3x1 + 5x2 = 20$, when $x2= 0$

$3x1 =20$

Then, $x1= 20/3$

Hence, $x1= 6.67$

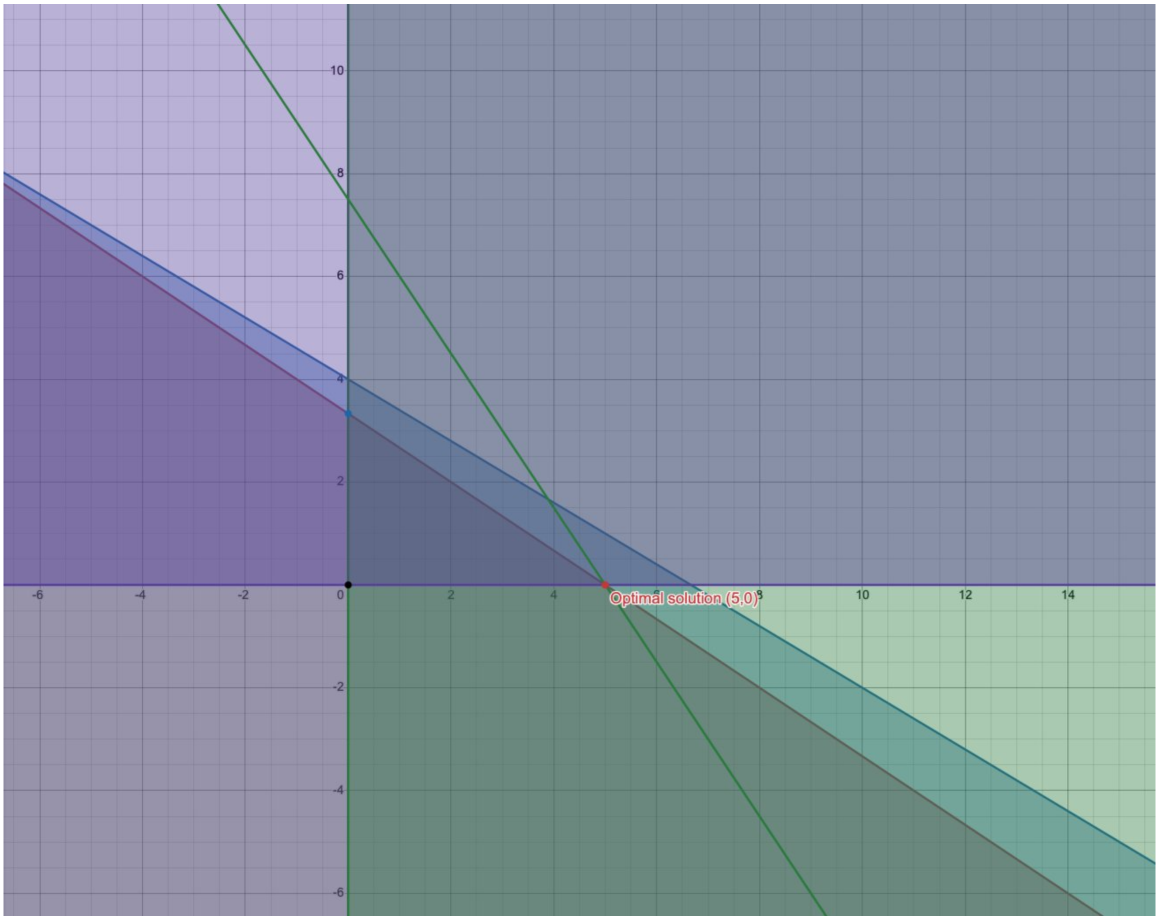
$3x1 + 5x2 = 20$, when $x1 = 0$

$5x2 = 20$

Then $x2= 20/5$

Hence, $x2= 4$

Corner	X1	X2	Z= 15x1+10x2
A	0	0	0
B	5	0	75
C	0	3.33	33.33



References:

- Caro, Felipe, Jérémie Gallien, Miguel Díaz, et al. 2010. ‘Zara Uses Operations Research to Reengineer Its Global Distribution Process’. *Interfaces* 40 (1): 71–84. https://doi.org/10.1287/inte.1090.0472.

- Energy, Hitachi. 2026. ‘AMPL for Energy: Optimization Solutions for a Mission-Critical Industry - AMPL’. AMPL, June 19. https://ampl.com/amplforenergy/.